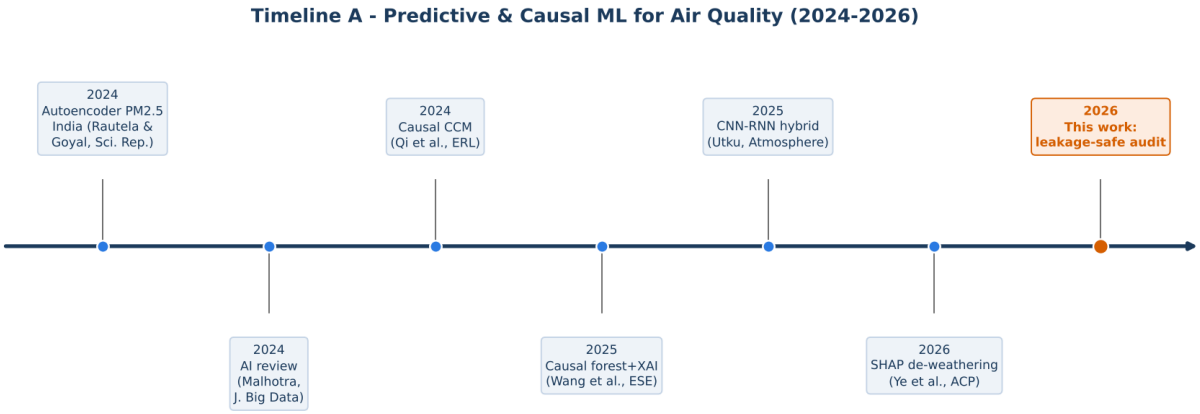
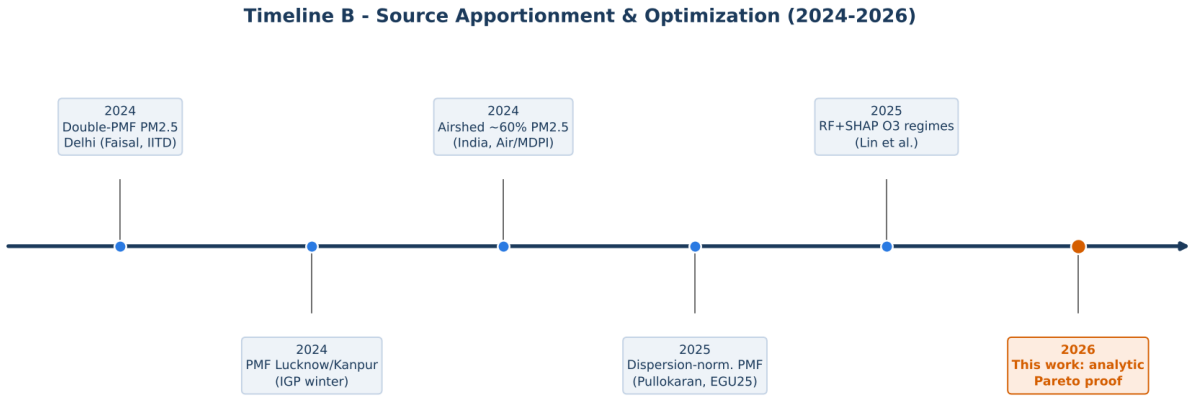
MDP Project

Literature Review & Research Gaps

Two timelines • gaps we solve • contributions & novelty • SCI/Q1 reading list



Timeline A - predictive & causal ML for air quality.



Timeline B - source apportionment & multi-objective optimization.

Literature Review, Timelines, Gaps, Contributions & Novelty

Prepared against the Review-2 checklist: *introduction gaps, research paper on a timeline (two timelines), the gaps we are solving, and our contributions & novelty*, grounded in SCI / Q1 literature.

> ***Scope of the timelines:*** *per mam's instruction, the two timelines below show only the* ***latest work (past ~2 years, 2024-2026)*** *-- not the older foundational papers. The field has two parallel recent threads:* ***Timeline A*** *= predictive & causal machine learning for air quality;* ***Timeline B*** *= source apportionment & multi-objective policy optimization. Our work sits at the end of both. (Foundational method papers are listed separately under §5 as background.)*

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1. Timeline A -- Predictive & Causal ML for Air Quality (2024-2026)

Year	Work (author, venue)	Contribution	Limitation it leaves
2024	Rautela & Goyal, *Scientific Reports* (Q1)	Convolutional autoencoder deriving PM2.5 across India	Accuracy-first; no causal/leakage analysis
2024	Malhotra et al., *Journal of Big Data* (Q1)	Systematic review of AI for air-pollution prediction	Confirms interpretability/evaluation gaps
2024	Qi et al., *Environ. Research Letters* (Q1)	Convergent cross-mapping for PM2.5 <-> O3 causality (China)	City/season-specific; not identification
2025	Wang et al., *Environ. Science & Ecotechnology* (Q1)	Causal forest + XAI for the 2022 ozone rebound (meteorology ~57%)	Observational; confounding only partly handled
2025	Utku et al., *Atmosphere* (Q2)	CNN-RNN hybrid PM2.5 forecasting (India + EU cities)	Random-style evaluation; no spatial transfer test
2026	Ye et al., *Atmos. Chem. Phys.* (Q1)	SHAP de-weathering of ozone trends (E. China)	Flags SHAP instability under multicollinearity
2026	**This work**	**Leakage-safe India benchmark + causal-stability audit + identification gate**	--

2. Timeline B -- Source Apportionment & Multi-Objective Optimization (2024-2026)

Year	Work (author, venue)	Contribution	Limitation it leaves
2024	Faisal (IIT Delhi, thesis)	Double-PMF / tracer-conjugate PMF of PM2.5, Delhi	Needs organic+elemental speciation
2024	Lakra et al., *Atmos. Environ.*	PMF on composite PM2.5 (Lucknow 8, Kanpur 7 factors)	Site/season specific; heavy data needs
2024	Nationwide airshed study, *Air* (MDPI)	Residential cooking/power/industry ~60% of India PM2.5	Inventory-based, not receptor at scale
2025	Pullokaran et al., *EGU25* (abstract)	Dispersion-normalised PMF across India	Conference abstract; speciation required
2025	Lin et al., *EGUsphere* (preprint)	RF + SHAP ozone-regime mapping via HCHO/NO2	Needs satellite/VOC proxies; associative
2026	**This work**	**Analytic Pareto proof: 100% NSGA-II dominance; concentration-emission fallacy**	--

> ***Note on the optimization half of Timeline B:*** *rigorous, recent (2024-2025) multi-objective optimization for air-quality policy is scarce -- the nearest applied work (e.g. a 2023 CNN-LSTM + multi-objective emission model over Xi'an stations) is already outside the two-year window and does not benchmark against an exact frontier.*

That scarcity is itself part of the gap our analytical Pareto result addresses.

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3. Research Gaps (what the recent literature leaves open -- and we address)

- **G1 -- Leakage-unaware evaluation.** Recent AQ-ML (e.g. forecasting papers in §1) still report random-style skill on autocorrelated data, overstating generalisation.
- **G2 -- Correlation read as mechanism.** SHAP/importance and discovered graphs are read causally; even 2026 work (Ye et al.) flags SHAP instability under multicollinearity, yet attribution is still treated as explanation.
- **G3 -- Untested meteorological / boundary-layer confounding.** Causal graphs among pollutants rarely test whether edges survive meteorology (the dominant confounder); recent causal work (Wang et al. 2025) targets this but only partially.
- **G4 -- Clustering mislabelled as source apportionment.** Correlation clustering is presented as apportionment despite lacking the speciation and uncertainties that PMF (and its 2024-25 dispersion-normalised variants) require.
- **G5 -- Concentration-emission conflation.** "Policy optimisation" acts on ambient concentrations, not emissions, and metaheuristics are not benchmarked against exact / analytic Pareto fronts -- a near-absent practice in the recent literature.
- **G6 -- Weak provenance QC and no open India benchmark.** Measurement-unit provenance is seldom audited, and there is no open, leakage-safe, reproducible multi-pollutant benchmark for the Indian network.

4. Our Contributions & Novelty (mapped to the gaps)

#	Contribution	Addresses	Novelty
C1	Leakage-safe, multi-protocol evaluation (random vs blocked-temporal vs leave-stations-out) with a full baseline ladder, on 204 Indian stations / 1.18M station-hours	G1	First India-scale, leakage-safe multi-pollutant benchmark
C2	Causal-discovery with bootstrap edge stability **and** an explicit meteorology-confounding test; only NO -> O3 survives	G2, G3	Couples stability + confounding testing rather than reporting a raw graph
C3	Formal distinction (and proof) that k-means correlation clustering is **not** PMF source apportionment	G4	Makes the clustering/apportionment boundary explicit and testable
C4	Analytical proof that the linear surrogate's Pareto front collapses to a 1-D segment; 100% of NSGA-II policies dominated (GD = 0.084, HVE = 6.6%)	G5	First closed-form optimality audit of a heuristic AQ optimiser
C5	Formalised **concentration-emission fallacy** (negative-mass reductio)	G5	Names a common category error in data-driven AQ policy
C6	Authoritative unit-provenance audit (ppb/µg-m-3 mix) + open reproducible benchmark + a 4-part **scientific identification gate**	G2-G6	A reusable gate for observational environmental AI

One-sentence novelty statement. To our knowledge this is the first India-scale, reproducibility-first audit that (i) proves analytically that the heuristic optimiser is dominated on the fitted surrogate, (ii) couples causal-discovery stability with meteorological-confounding tests, and (iii) formalises a scientific identification gate for observational environmental AI.

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5. SCI / Q1 Reading List

Confirm the current quartile on Scimago (SJR) or JCR before citing -- rankings change yearly and by subject category.

A. Recent, peer-reviewed SCI/Q1 (read these first -- past ~2 years) DOIs verified 2026-10-31 (resolve each at https://doi.org/<DOI>); quartiles are indicative -- confirm on Scimago for the exact subject category.

1. Rautela, K. S. & Goyal, M. K. (2024). *Transforming air pollution management in India with AI and machine learning technologies.* **Scientific Reports** 14, 20412. DOI 10.1038/s41598-024-71269-7. (Q1)

2. Malhotra, M., Walia, S., Lin, C.-C., Aulakh, I. K. & Agarwal, S. (2024). *A systematic scrutiny of artificial intelligence-based air pollution prediction techniques, challenges, and viable solutions.* **Journal of Big Data** 11, 142. DOI 10.1186/s40537-024-01002-8. (Q1)

3. Qi, L., Yin, J., Li, J. & Duan, X. (2024). *Detecting causal relationships between fine particles and ozone based on observations in four typical cities of China.* **Environmental Research Letters** 19 (Jan 2024). **DOI: confirm on IOPscience** (could not be auto-resolved here; paper confirmed to exist). (Q1) **Read for G2/G3.**

4. Wang, L., Chen, B. et al. (2025). *Causal-inference machine learning reveals the drivers of China's 2022 ozone rebound.* **Environmental Science and Ecotechnology** 24, 100524. DOI 10.1016/j.ese.2025.100524. (Q1) **Read for G2/G3.**

5. Utku, A. et al. (2025). *Advancing Air Quality Monitoring: Deep Learning-Based CNN-RNN Hybrid Model for PM2.5 Forecasting.* **Atmosphere** 16(9), 1003. DOI 10.3390/atmos16091003. (Q2)

6. Ye, X., Zhang, L., Wang, X., Lu, N., Hickman, S., Luo, G. & Archibald, A. T. (2026). *Deciphering the impacts of meteorology on surface ozone variability in eastern China using explainable machine learning models.* **Atmospheric Chemistry and Physics** 26(9), 6377. DOI 10.5194/acp-26-6377-2026. (Q1) **Read for G2.**

7. Lakra, K., Shukla, T., Bhowmik, H. S. et al. (2024). *Comparative analysis of winter composite-PM2.5 in Central Indo-Gangetic Plain cities.* **Atmospheric Environment** 338, 120827. DOI 10.1016/j.atmosenv.2024.120827. (Q1)

B. Recent grey literature (very current; not yet peer-reviewed -- cite with care)

8. Lin et al. (2025), *EGUsphere* preprint -- RF+SHAP ozone-regime mapping (HCHO/NO2).

9. Pullokaran et al. (2025), *EGU25* abstract -- dispersion-normalised PMF across India.

10. Faisal (2024), IIT Delhi thesis -- double-PMF / tracer-conjugate PMF, Delhi.

C. Foundational methods (background -- cite for method definitions, not as "recent")

11. Paatero & Tapper (1994), *Environmetrics* (Q1) -- PMF. DOI 10.1002/env.3170050203

12. Deb et al. (2002), *IEEE Trans. Evolutionary Computation* (Q1) -- NSGA-II. DOI 10.1109/4235.996017

13. Shimizu et al. (2011), *J. Machine Learning Research* (Q1) -- DirectLiNGAM.

14. Lundberg & Lee (2017), *NeurIPS*; Lundberg et al. (2020), *Nature Machine Intelligence* (Q1) -- SHAP.

15. Just et al. (2020), *Atmospheric Environment* 239:117649 (Q1) -- spatial CV vs leakage in PM2.5 ML. DOI 10.1016/j.atmosenv.2020.117649. **Read for G1.**

16. Grange & Carslaw (2019), *Sci. Total Environment* (Q1) -- meteorological normalisation.

17. Sillman (1999), *Atmospheric Environment* (Q1) -- VOC- vs NOx-limited ozone.

18. Seinfeld & Pandis (2016), textbook (Wiley) -- reference atmospheric chemistry.

How to split the reading (5 students)

- **Member 1 -- recent prediction/DL:** #1, #2, #5
- **Member 2 -- recent causal & XAI:** #3, #4, #6, #8
- **Member 3 -- recent apportionment/PMF:** #7, #9, #10, #11
- **Member 4 -- leakage & evaluation:** #15, #16
- **Member 5 -- optimization & chemistry:** #12, #17, and one recent applied optimization paper

Each reader: note the paper's method, its **evaluation protocol**, and which of our gaps (G1-G6) it supports or challenges -- that mapping is exactly what strengthens our Introduction.